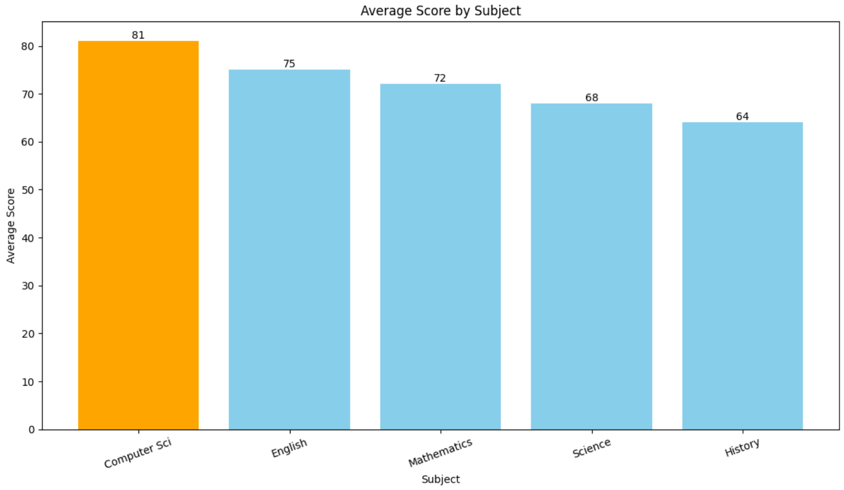
Question 1

Scenario	Best Chart Type	Justification
1. Comparing total revenue across 6 product categories in one year.	**Bar Chart (Amounts)**	A bar chart is best for comparing numerical values across different categories. It clearly shows which product category has the highest and lowest revenue.
2. Showing how the salary distribution differs across 4 job levels.	**Box Plot (Distribution)**	A box plot displays the distribution, median, spread, and outliers of salaries, making it easy to compare salary distributions across job levels.
3. Showing market share of 5 competing brands.	**Pie Chart (Proportion)**	A pie chart effectively shows how each brand contributes to the total market share as percentages.
4. Showing the relationship between advertising spend and sales revenue.	**Scatter Plot (X–Y Relationship)**	A scatter plot helps identify the relationship or correlation between advertising spending and sales revenue.
5. Showing monthly website traffic over 3 years.	**Line Chart (Temporal)**	A line chart is ideal for displaying trends and changes in website traffic over time.
6. Showing which employees report to which managers in an organization.	**Network Diagram (Network)**	A network diagram clearly represents reporting relationships between employees and managers using connected nodes.

Question 2 – Vertical Bar Plot

```
import pandas as pd

import matplotlib.pyplot as plt

# Create DataFrame

df = pd.DataFrame({

    'Subject': ['Mathematics', 'Science', 'English', 'History', 'Computer Sci'],

    'Average_Score': [72, 68, 75, 64, 81]

})
```

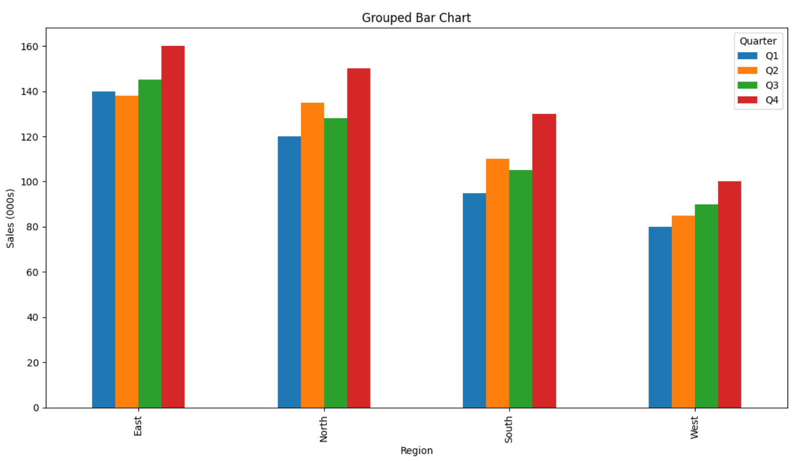
```python
# Sort descending
df = df.sort_values(by='Average_Score', ascending=False)
# Colors
colors = ['orange' if x == df['Average_Score'].max() else 'skyblue'
          for x in df['Average_Score']]
# Plot
plt.figure(figsize=(8,5))
bars = plt.bar(df['Subject'], df['Average_Score'], color=colors)
# Labels
for bar in bars:
    plt.text(bar.get_x()+bar.get_width()/2,
             bar.get_height()+0.5,
             f'{bar.get_height()}',
             ha='center')
plt.title("Average Score by Subject")
plt.xlabel("Subject")
plt.ylabel("Average Score")
plt.xticks(rotation=20)
plt.show()
```



Question 3

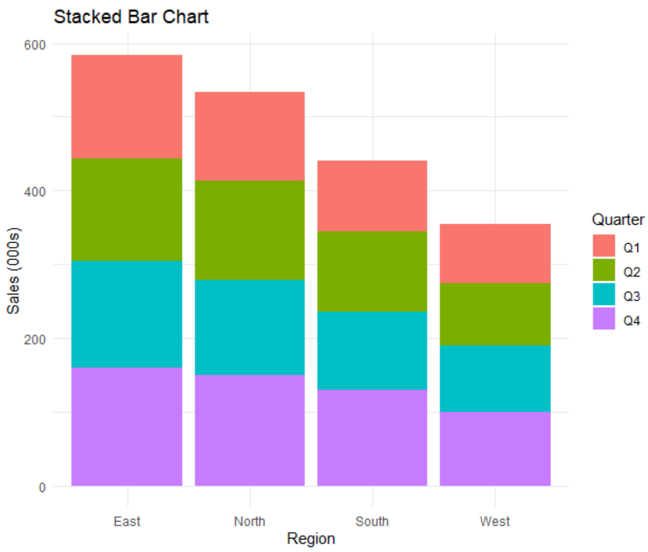
1. Grouped Bar Chart

```
import pandas as pd

import matplotlib.pyplot as plt

data = {

'Region':['North','North','North','North',

         'South','South','South','South',

         'East','East','East','East',

         'West','West','West','West'],

'Quarter':['Q1','Q2','Q3','Q4']*4,

'Sales_000s':[120,135,128,150,

             95,110,105,130,

             140,138,145,160,

             80,85,90,100]

}

df = pd.DataFrame(data)

pivot = df.pivot(index='Region', columns='Quarter', values='Sales_000s')

pivot.plot(kind='bar', figsize=(8,5))

plt.title("Grouped Bar Chart")

plt.ylabel("Sales (000s)")

plt.show()
```



2. Stacked Bar Chart

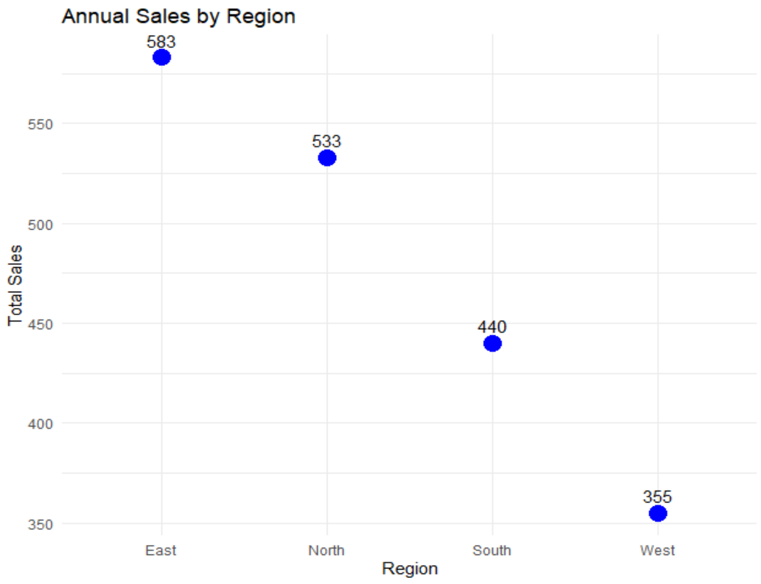
```
ggplot(df, aes(x = Region, y = Sales_000s, fill = Quarter)) +
  geom_col(position = "stack") +
  labs(
    title = "Stacked Bar Chart",
    x = "Region",
    y = "Sales (000s)"
  ) +
  theme_minimal()
```



2. Dot Plot (Annual Sales by Region)

```
library(ggplot2)
# Dataset
df <- data.frame(
  Region = rep(c("North", "South", "East", "West"), each = 4),
  Quarter = rep(c("Q1", "Q2", "Q3", "Q4"), 4),
  Sales_000s = c(
    120,135,128,150,
```

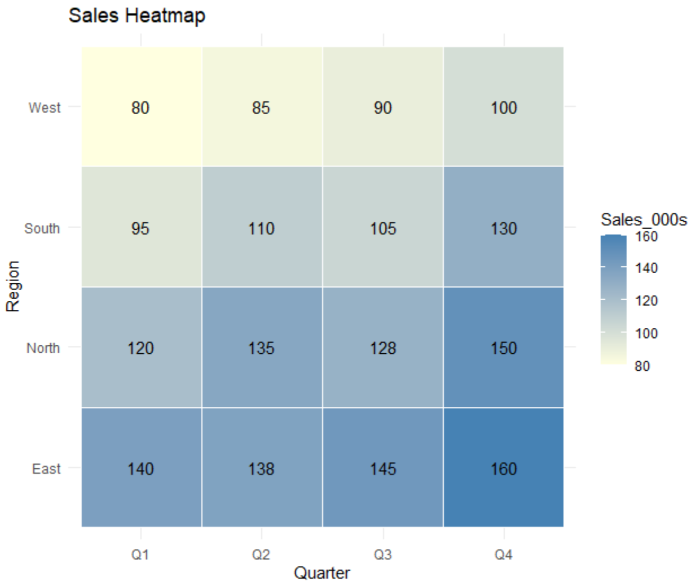
```
    95,110,105,130,
    140,138,145,160,
    80,85,90,100
  )
)
# Calculate total sales
total <- aggregate(Sales_000s ~ Region, data = df, sum)
# Rename column
colnames(total)[2] <- "Total"
# Dot Plot
ggplot(total, aes(x = Region, y = Total)) +
  geom_point(size = 5, color = "blue") +
  geom_text(aes(label = Total), vjust = -0.8) +
  labs(title = "Annual Sales by Region",
       x = "Region",
       y = "Total Sales") +
  theme_minimal()
```



4. Heatmap

```
ggplot(df, aes(x = Quarter, y = Region, fill = Sales_000s)) +
```

```
geom_tile(color = "white") +
geom_text(aes(label = Sales_000s), color = "black") +
scale_fill_gradient(low = "lightyellow", high = "steelblue") +
labs(
  title = "Sales Heatmap",
  x = "Quarter",
  y = "Region"
) +
theme_minimal()
```

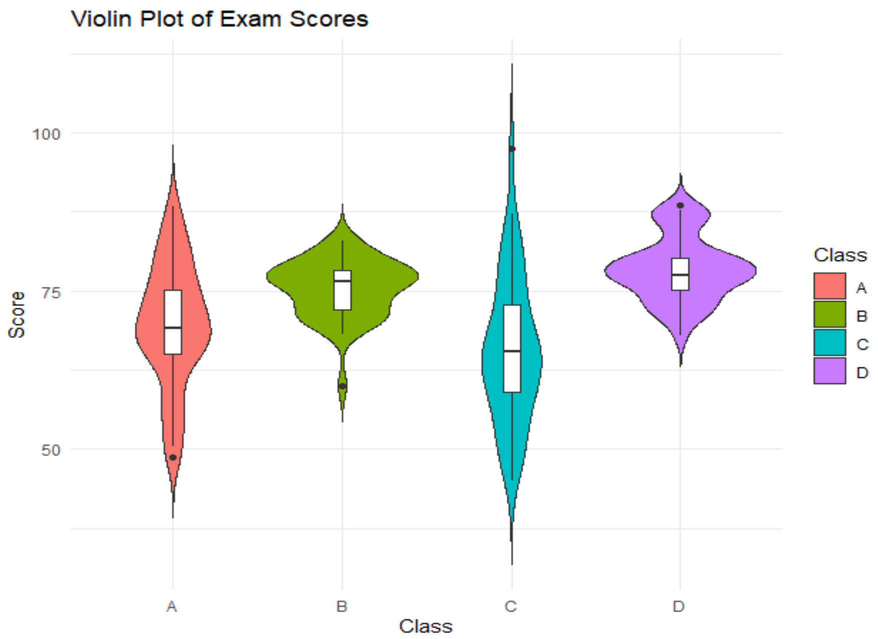


Question 4

1. Violin Plot with Boxplot

```
ggplot(scores, aes(x = Class, y = Score, fill = Class)) +
  geom_violin(trim = FALSE) +
  geom_boxplot(width = 0.1, fill = "white") +
  labs(
    title = "Violin Plot of Exam Scores",
    x = "Class",
    y = "Score"
  ) +
```

theme_minimal()



2. Ridgeline Plot

ggplot(scores,

aes(x = Score, y = Class, fill = Class)) +

geom_density_ridges(alpha = 0.7) +

labs(

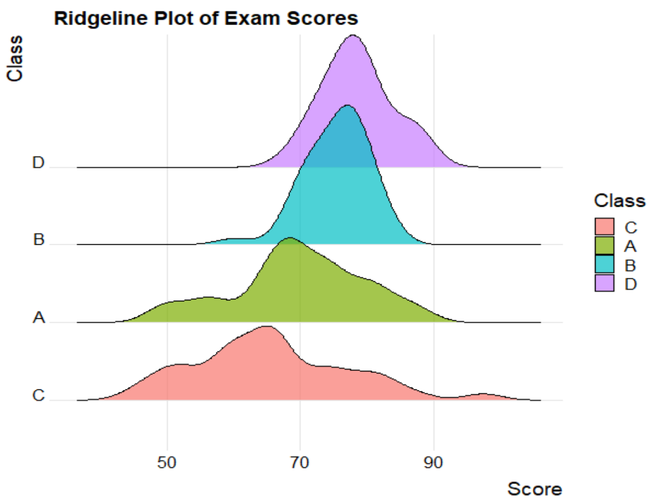
title = "Ridgeline Plot of Exam Scores",

x = "Score",

y = "Class"

) +

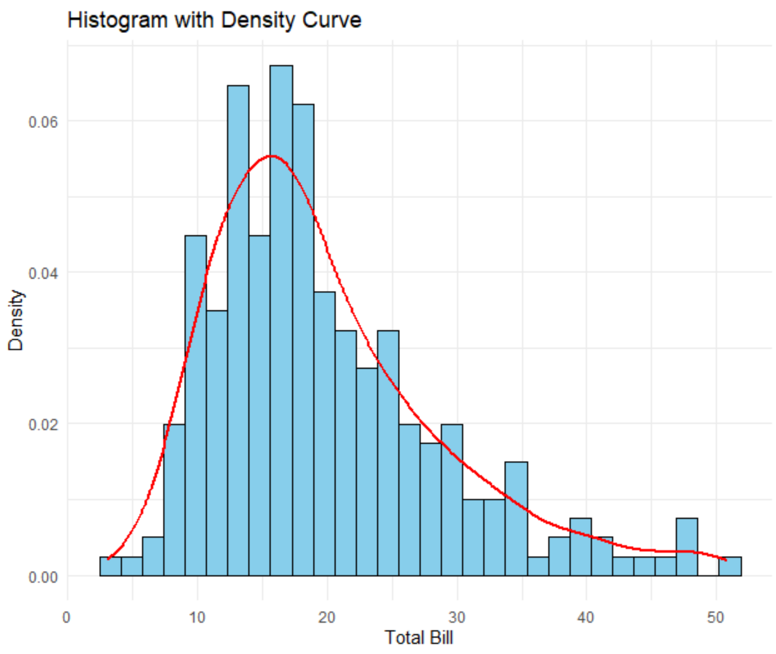
theme_ridges()



Question 5

1. Histogram with KDE

```
ggplot(tips, aes(x = total_bill)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 30,
                 fill = "skyblue",
                 color = "black") +
  geom_density(color = "red", linewidth = 1) +
  labs(
    title = "Histogram with Density Curve",
    x = "Total Bill",
    y = "Density"
  ) +
  theme_minimal()
```



2. Histogram + KDE by Smoker

```
ggplot(tips,
 aes(x = total_bill,
     fill = smoker,
     color = smoker)) +
geom_histogram(aes(y = after_stat(density)),
               bins = 30,
               alpha = 0.5,
               position = "identity") +
geom_density(linewidth = 1) +
labs(
  title = "Histogram and Density by Smoker",
  x = "Total Bill",
  y = "Density"
) +
theme_minimal()
```